Behavioural Brain Research 196 (2009) 15–29

Contents lists available at ScienceDirect

Behavioural Brain Research

journal homepage: www.elsevier.com/locate/bbr

Research report

Hand preferences for bimanual coordination in 29 bonobos (*Pan paniscus*)

Amandine S. Chapelain*, Eef Hogervorst

Department of Human Sciences, Loughborough University, Loughborough, Leicestershire LE11 3TU, UK

ARTICLE INFO

Article history:
Received 29 February 2008
Received in revised form 26 June 2008
Accepted 1 July 2008
Available online 18 July 2008

Keywords:
Brain lateralization
Hand preferences
Bimanual coordination
Bonobos

ABSTRACT

Brain lateralization has long been thought to be unique to humans. To investigate the origins and functions of this feature, researchers study behavioural laterality in other animals. Despite a substantial database, manual laterality in non-human primates remains a controversial topic. We give here a review of the main findings on manual preference in great apes. This article presents data on hand preferences for a bimanual coordination in 29 bonobos (*Pan paniscus*). The study aims to provide data on manual laterality for a complex bimanual task in this very interesting and rarely studied species. Hand preferences were assessed using the 'tube task'. This task has been used with other species, which allows reliable data comparisons. The task requires a bimanual coordinated precise action: the subject holds the tube with one hand while reaching for food inside with the other hand. As a complex task, this measure has been shown to be efficient in revealing hand preferences. It has revealed group-level right bias in chimpanzees. Bonobos had never been tested. We recorded both independent bouts (counting only the first pattern of a sequence of identical actions) and frequency (counting every action). The bonobos exhibited strong hand preferences. With frequency, 11 bonobos were classified as right-handed, 15 were left-handed and 3 had no preference. With bouts, 8 bonobos were right-handed, 9 were left-handed and 12 had no preference. No group-level bias appeared. The results are discussed in relation with previous findings and theories on brain lateralization.

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1. Introduction

Brain lateralization has long been thought to be unique to humans, and related to human-specific traits such as language or handedness. In humans, 90% of the individuals prefer to use the right hand for complex tasks [3,25]. This population-level right-handedness is constant around the 90% mark, with little variation in the proportion of left-handers between populations (from 0% to 27%) [19,25,30,31,76,96]. This preference for using the right hand would reflect a specialization of the left hemisphere for manual functions [17,25]. Despite extensive research, the origins and functions of right-handedness in humans remain an unsolved issue [25,30].

The function of brain lateralization would be to improve brain efficiency [115,116]. It would provide several advantages, such as saving neural space by avoiding replication of functions, allowing simultaneous processing of different processes and avoiding hemispheric competition [17,97]. There is evidence that brain lateralization would improve cerebral abilities as well as behavioural efficiency (in humans, non-human primates and other animals [21,30,35,80,81,98,100,108]). Along with that, the brain should be lateralized for the complex processes that are important for survival, and behavioural laterality should appear in such tasks [25,26].

Assuming that behavioural preferences (laterality) reflect brain lateralization, hand preference is used as an indicator of brain lateralization for manual functions [17]. To investigate the evolution of human brain lateralization and handedness, researchers study hand preference in other primates. Non-human primates data can provide valuable clues to understand when handedness would have appeared, and what could have been the selective factors for its emergence: e.g. postural demands [84], tool use [68], throwing [9], bipedalism, gestural communication and language [18,40,117], complex manipulations [6], precise actions, bimanual coordination [26] (theories reviewed in [79,92]).

Hand preference has been extensively studied in non-human primates, but the results have often been contradictory and inconsistent. Therefore, and despite a substantial database, this topic has long been and remains controversial [26,47,61,70,79,84,90,92,122]. In a nutshell, studies have shown that non-human primates exhibit hand preferences. However, the pattern observed in these primates would differ from the human pattern. In non-human primates, individual preferences have been found for several tasks. But the

* Corresponding author. Tel.: +44 1509 223048/33 3 81 58 55 84; fax: +44 1509 223940.
E-mail addresses: a.s.chapelain@lboro.ac.uk, chapelain_amandine@yahoo.fr (A.S. Chapelain).

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doi:10.1016/j.bbr.2008.07.012

preferences tend to be relatively weak, and they vary between tasks. At the group-level, the numbers of right-handers and left-handers are about identical. Evidence of group-level biases is scarce and disputed, and no human-like group-level bias has been reported (data reviewed in [26,47,61,70,79,84,92,122]).

However, there are important methodological problems with the assessment of hand preference in non-human primates [26,61,72,79]. The first one is related to task specificity. Research has shown that the strength and direction of preference vary according to the task (e.g. [2,13,35,58,75,92,109,122]). This raises issues because the results thus depend on the task that is used to assess laterality. It would be essential to standardize the methods between studies to allow comparisons of the data between studies and species. Disappointingly, there is no consensus between studies regarding these measures [47,61]. Thus, when considering the available database, it is difficult to get an understandable overview of manual laterality in non-human primates and to interpret the findings. This problem also applies when comparing studies on humans and studies on non-human primates [61,106]. The measures used to assess laterality are different and this might account for some difference in the results.

We first present a review of the data on manual laterality in great apes. Great apes are a very interesting model for investigating the evolution of human brain lateralization. They are close to humans on several important aspects. Firstly, they are close in the phylogeny. It is unlikely that brain lateralization appeared *de novo* in humans, and so, precursors may have been present earlier in the human-ape lineage. Moreover, great apes display outstanding characteristics in terms of brain lateralization and handedness. Their hands are similar to those of humans regarding morphology and manipulative skills [8]. They can manufacture and use tools (e.g. [82]). They can use bipedal locomotion (e.g. [24,113]). They exhibit some abilities for language (e.g. [103]). All of these features have been proposed as mechanisms fostering the evolution of brain lateralization and handedness. Finally, great apes would exhibit some neuroanatomical brain asymmetries that might be close to those observed in humans [10,36,41,52]. Along with that, the great apes would be expected to exhibit a brain lateralization and behavioural laterality that could be close to the one of humans.

Table 1 presents the main data on hand preference in great apes. In summary, these studies indicate that the great apes exhibit hand preferences. Laterality has been found to be weak for simple tasks such as tripedal reaching, but strong for complex tasks that require high cognitive skills. For instance, for bimanual coordinated actions and for tool use, laterality was very pronounced, individual preferences were very strong and most of the individuals were lateralized. At the group-level, biases have been found only for certain behaviours (e.g. bimanual feeding, bimanual 'tube task', bipedal reaching, gesturing) and they never reached a human-like handedness bias.

1.1. Model of study: bonobos (*Pan paniscus*)

Data on hand preference in great apes are crucial to research on the evolution of human handedness. Amongst great apes, bonobos are particularly close to humans and interesting. *Pan* is the closest genus to humans in the phylogeny [69], which makes bonobos great candidates to exhibit precursors of human features. Bonobos have linguistic abilities that could surpass those of the other apes [102,103]. Both language and handedness are lateralized in the left hemisphere in humans [25]. There are several theories proposing a relationship between the evolution of language and handedness lateralizations [18,40,117]. Bonobos would be the non-human primates that most frequently use bipedal locomotion [24,112,113]. Bipedalism has been proposed to be related to the emergence of right-handedness in humans. Therefore, data on hand preference in bonobos are of great interest for discussing these evolutionary theories. The theories predict that brain lateralization in bonobos should be close to the one of humans. Regarding behavioural laterality, bonobos would be expected to exhibit strong individual preferences and a group-level right bias.There has been very little investigation of manual laterality in bonobos. There are fewer than ten published studies [47,72]. And these considered only small groups (N between 2 and 14) (except [37] $N = 22$). They assessed hand preferences in spontaneous daily activities (e.g. feeding, carrying object/infant, leading limb, reaching, manipulating, self touching, gesturing) [37,44,48,67,107] and in experimental tasks (reaching from different postures) [14,16,23,44]. This would indicate that the brain of bonobos may be lateralized for manual functions. Their findings indicate that bonobos exhibit manual preferences. The preferences were present at the individual-level (except group-level biases reported for leading limb in locomotion (right bias) and carrying (left bias) with combined data from [44,48]).

1.2. Task used to assess laterality

We study hand preference in bonobos to investigate the evolution of human brain lateralization and handedness. Comparing the pattern of hand preference between humans and non-human species can potentially bring essential light to the discussion. Unfortunately, the comparisons are difficult because of a lack of standardization in the methods used to assess laterality. The measures used with humans and those used with non-human primates are different [61,106]. The tasks used with humans are complex (e.g. tool use, complex precise actions) [25], while most tasks used with non-human primates are simpler, i.e. unimanual non-sequential and unprecise actions [72]. When humans are tested with simple tasks, similar to those used with non-human primates, their laterality is less pronounced (e.g. [3,30,76,106,110]). To determine whether non-human primates could display human-like handedness it would be necessary to test them with complex tasks.

Researchers agree that laterality would be more easily revealed in complex than simple tasks. According to the "theory of task complexity" [26], strong preferences and group-level biases would be more likely to appear in complex tasks. This has been consistently reported and is generally admitted: the more complex the task is, the stronger the laterality is (humans: e.g. [25,38,76,110], non-human primates: e.g. [2,6,15,27,28,35,95,104,109,114,118,122]). Task complexity can be related to different aspects of the task (e.g. novelty, precision, manipulation, bimanual coordination, sequence of actions, subordinate hand used for postural support, complex balance control) that make the task cognitively demanding [26,79].

Specifically, a precise bimanual manipulation on one object is thought to be the best task to require and evoke laterality [7,8,47,99]. In a bimanual manipulative task, the two hands are engaged in a coordinated action, and they have different but complementary roles (manipulate the item/hold the item) [20]. Such tasks would be efficient measures to reveal hand preference [47]. Moreover, these tasks are less subjected to the influence of situational factors [54] (e.g. object position [29,74,106,119]), and manual preference is thus more likely to reflect brain lateralization [47,53,54].

Therefore, we have chosen a coordinated bimanual task to measure hand preference in bonobos. The task requires the use of both hands in a coordinated way along with a precise action: the individual holds a tube with one hand while reaching for food inside with a finger of the other hand [55]. This 'tube task' has been used in other studies, and may become the long-awaited standard measure of hand preference in non-human primates. Table 2 and Fig. 1

Table 1
Review of the main data on hand preference in great apes

Task	Species	Observations
Feeding: hand that brings food to the mouth	Chimpanzees (wild)	Individual preferences [75]
	Bonobos	Individual preferences [37,44,107]
	Gorillas (wild)	Individual preferences [93]
	Orang-utans (captive and rehabilitated)	Individual preferences [89,99]
Bimanual feeding: one hand brings food to the mouth while the other hand holds another food item	Chimpanzees	Group-level right bias [54]
	Bonobos	Individual preferences. Trend toward right hand use [44,48]
	Orang-utans (wild and captive)	Individual preferences in [94], no preference in [89]
Bimanual coordinated manipulative action: the two hands are engaged in a coordinated action, with different but complementary roles		
: one hand manipulates the item while the other hand holds the item	Chimpanzees (wild)	For bimanual processing of *Saba* fruit and lemons: very strong individual preferences (often exclusive use of one hand). Almost all of the individuals lateralized [7,20]
-On food	Gorillas (wild)	For bimanual sequential manipulation of plants with defenses: very strong preferences (often exclusive use of one hand). Majority of the individuals lateralized. Trend towards a right hand preference for fine manipulation while the left hand holds the item [6]
	Orang-utans (rehabilitated)	Strong individual preferences [99]
-On an object	Chimpanzees, gorillas, orang-utans	For the 'tube task': Table 2
Manipulative tasks requiring to open a box or slide a panel	Chimpanzees	Individual preferences. Most individuals lateralized [15]
	Gorillas	Individual preferences. Most individuals lateralized [28,88]
	Orang-utans	Individual preferences. Most individuals lateralized [88]
Tripedal reaching: reaching for items on the ground from a tripedal posture	Chimpanzees	Few individuals lateralized [16,32,45,58,74]
	Bonobos	Few individuals lateralized [23,44,48]
	Gorillas	Few individuals lateralized [28,88]
	Orang-utans	Few individuals lateralized [39,45,88]
Bipedal reaching: reaching for items placed at height from a bipedal posture	Chimpanzees	Group-level right bias [45]
	Bonobos	Trend toward right hand use [44]. Enhanced right hand use in bipedal compared to tripedal posture [44] but reverse in [23]
	Gorillas	Group-level right bias [88]
	Orang-utans	Individual preferences [45,88]
Throwing	Chimpanzees	Strong laterality [16,50]. Individual preferences in [16,73], group-level right trend in [50], group-level right bias in [51]
Tool use	Chimpanzees (wild)	Very strong individual preferences (almost always exclusive use of one hand). Most of the individuals lateralized: to use a probe to fish into termite nests [71,75,81,80,87]. (a group-level left trend in [71]), to use a sponge to drink [5], to use a stone to crack open nuts [5,77,111], to pound *strychnos* fruits [83]
	Chimpanzees	To use a probe to fish for yoghurt into an artificial termite mound: half of the individuals lateralized. Group-level left bias [34].
	Bonobos	Individual preferences (Chapelain unpublished data) [37,107]
	Orang-utans	To make/modify tools and to use tools: individual preferences. Strongest laterality relative to other tasks [89]
Gestures	Chimpanzees	For induced gestures: begging for food from the experimenter: group-level right bias [57,59,65] (right preference more pronounced for gestures accompanied by a vocalization [43]) For spontaneous gestures: clapping hands together to attract the attention of humans: very strong individual preferences (quasi exclusive preferences). Almost all individuals lateralized [33]
	Bonobos	Spontaneous gesturing (intraspecific and interspecific): weak laterality. Most individuals without preference [37,44,48,107], some strong individual preferences in [11]
Cradling	Chimpanzees and gorillas	Some evidence of group-level Left bias (reviews in [22,46])
Carrying	Chimpanzees	Individual preferences [34,73]
	Bonobos	Group-level left bias (with combined data from [44,48])

Task: behaviour considered. Species: great ape species considered. These are captive animals, except when specified wild or rehabilitated. Observations: main findings. Individual preferences or group-level bias. Group-level biases have been calculated on the number of right-handed and left-handed individuals (binomial test).

Table 2
Review of the studies that used the 'tube task' to measure hand preferences

Study	Species	Location	Number of subjects (*N*)	Measure	Number of data points per subject	Number of test sessions	Right-handed	Left-handed	Not-lateralized	*B* test
Hopkins [55]	Chimpanzees, *Pan troglodytes*		110	Frequency	From 17 to 515	2 test sessions separated by several days	59	32	19	**0.006**
Hopkins [56]	Chimpanzees, *Pan troglodytes*		188	Frequency	Mini 20		95	53	40	**7E−04**
Hopkins et al. [53]	Chimpanzees, *Pan troglodytes*	Yerkes Research Center, Georgia	109 (some are the same as in [55])	Frequency	Mean 118.52	4 (2 sessions per day on 2 successive days)	54	33	22	**0.031**
Hopkins et al. [53]	Chimpanzees, *Pan troglodytes*	Yerkes Research Center, Georgia	109 (some are the same as in [55])	Bouts	Mean 23.78	4 (2 sessions per day on 2 successive days)	24	15	70	0.2
Hopkins et al. [64]	Chimpanzees, *Pan troglodytes*	Alamogordo primate facility, New Mexico	148	Frequency	Mini 20	At least 2 test sessions				
Hopkins et al. [64]		University of Texas, Bastrop	117	Frequency						
Hopkins et al. [64]		Yerkes Research Center, Georgia	202	Frequency						
Total for these 3 populations			467	Frequency			272	162	33	**1E−07**
Total for these 3 populations			452	Bouts			269	183	0	**6E−05**
Hopkins et al. [66]	Chimpanzees, *Pan troglodytes*	Yerkes Research Center, Georgia	180 (some are the same as in [53])	Bouts		8 (4 tests sessions per day on 2 days)	67	30	83	**2E−04**
Hopkins et al. [63]	Gorillas, *Gorilla gorilla*		31	Frequency		4 (2 sessions per day on 2 successive days)	15	12	4	0.701
Hopkins et al. [63]	Orang-utans, *Pongo pygmaeus*		19	Frequency		4 (2 sessions per day on 2 successive days)	4	15	0	**0.019**
Vauclair et al. [120]	Baboons, *Papio anubis*	Station de Rousset sur Arc, France	104	Frequency	From 15 to 311, mean 80		52	33	19	**0.05**
Meunier and Vauclair [85]	White-faced capuchins, *Cebus capucinus*	Centre de primatologie, Strasbourg, France	13	Frequency	Mini 50		6	6	1	1.226

Table 2 (*Continued*)

Study	Species	Location	Number of subjects (*N*)	Measure	Number of data points per subject	Number of test sessions	Right-handed	Left-handed	Not-lateralized	*B* test
Phillips and Sherwood [95]	Tufted capuchins, *Cebus apella*		7	Frequency	Mean 34 (S.D. 18)	4 test sessions separated by several days	3	3	1	1.313
Westergaard and Suomi [124]	Tufted capuchins, *Cebus apella*		45	Frequency	From 20 to 175, mean 81	2 test sessions separated by 1 to several days	19	20	6	1
Westergaard and Suomi [124]	Tufted capuchins, *Cebus apella*		18 (subsample of the 45 subjects)	Bouts	25		10	7	1	0.629
Westergaard and Suomi [124]	Rhesus macaques, *Macaca mulatta*		55	Frequency	From 20 to 128, mean 66	2 test sessions separated by one to several days	28	15	12	0.066
Westergaard et al. [125]	Infant rhesus macaques, *Macaca mulatta*		19	Frequency	From 20 to 50, mean 42	2 test sessions	3	12	4	**0.035**
Schweitzer et al. [104]	De Brazza's monkeys, *Cercopithecus neglectus*	Station de Paimpont, France	12	Frequency	From 82 to 279, mean 202.4	20 test sessions (1 session per day)	3	9	0	0.146
Chapelain and Hogervorst (this study)	Bonobos, *Pan paniscus*	Twycross zoo (UK), Stuttgart (Germany) zoo, Apenheul (Holland)	29	Frequency	From 47 to 656, mean 272	10 test sessions (1 session per day)	11	15	3	0.557
Chapelain and Hogervorst (this study)	Bonobos, *P. paniscus*	Twycross zoo (UK), Stuttgart (Germany) zoo, Apenheul (Holland)	29	Bouts	From 15 to 98, mean 47.62	10 test sessions (1 session per day)	8	10	11	0.815

Right-handed: number of right-handed subjects, left-handed: number of left-handed subjects, not-lateralized: number of not-lateralized subjects. B test: *p* value for the binomial test performed on the number of right-handed vs. left-handed subjects, significant values indicating group-level bias are in bold.

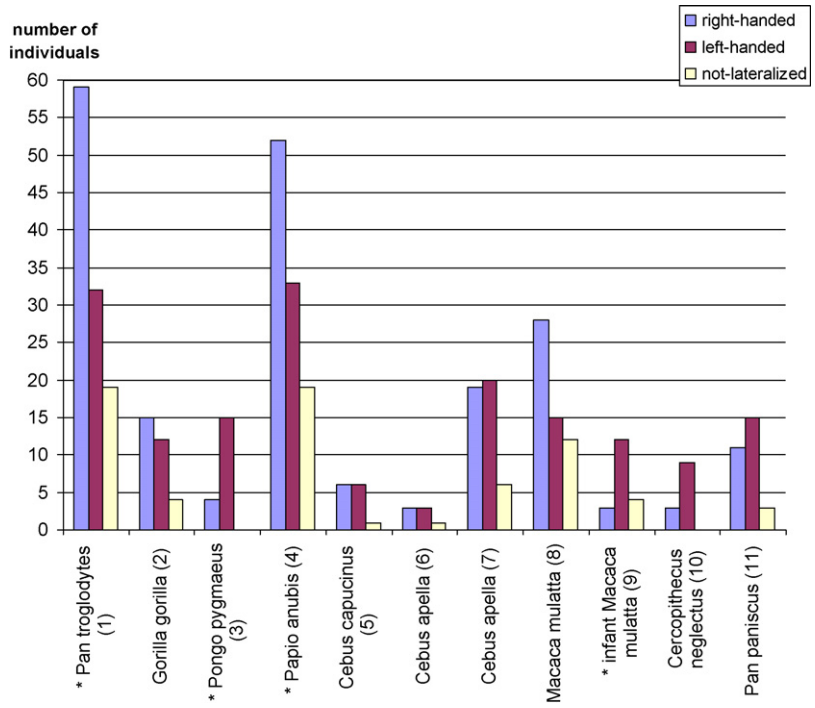


Fig. 1. Review of the data for species tested with the tube task (frequency). Number of individuals in each of the three categories (right-handed, left-handed, not-lateralized) for each study. The numbers in brackets correspond to the study: (1) Hopkins [55] (first of his studies on chimpanzees, other studies not represented here), (2) Hopkins et al. [63], (3) Hopkins et al. [63], (4) Vauclair et al. [120], (5) Meunier and Vauclair [85], (6) Phillips and Sherwood [95], (7) Westergaard and Suomi [124], (8) Westergaard and Suomi [124], (9) Westergaard et al. [125], (10) Schweitzer et al. [104], (11) Chapelain and Hogervorst (this study). Asterisks indicate group-level biases according to the binomial test performed on the number of right-handed vs. left-handed subjects.

present a review of these 'tube task' data. Previous studies indicate that the 'tube task' is a sensitive measure of hand preference [61,85,120,124]. It revealed a marked laterality, with strong individual preferences and most of the individuals being lateralized. Several group-level biases have been reported: a left bias in orang-utans [63], a left bias in infant rhesus macaques [125], a right bias in baboons [120] and a right bias in chimpanzees (replicated in several studies (reviewed in [61]). The preferences appeared to be stable across time (up to several years) hereby showing the reliability of this task [53,66,85,120,124]. Thus, the 'tube task' would be an appropriate measure to assess hand preference. This task has been used with the other great ape species: orang-utans ($N = 19$), gorillas ($N = 31$) [63] and a very large sample of chimpanzees ($N = 467$) [47], but bonobos had never been tested.

In this study, we assessed hand preferences in a sample of 29 bonobos using the 'tube task'. Our aims were to provide data on manual laterality for a complex bimanual task in this very interesting and rarely studied species. These data would be crucial to the study of hand preference in bonobos, and appropriate for comparisons with the other species. Along with the evolutionary theories and the "theory of task complexity" [26], bonobos would be expected to exhibit strong individual preferences and a group-level right bias for the 'tube task' [26].

2. Materials and methods

2.1. Subjects (Table 3)

The subjects were 29 bonobos (*P. paniscus*): 16 adults, 5 adolescents, 8 juveniles [4], ranging in age from 3 to 40 years. 21 bonobos were mother-reared and 8 were human-reared (nursery reared). They were housed in three zoos: 8 bonobos at Twycross zoo, Twycross (England), 13 bonobos separated in two groups at the Wilhelma zoo, Stuttgart (Germany) and 8 bonobos at Apenheul, Apeldoorn (Holland). The whole population tested included 35 bonobos, but we obtained enough data for 29 individuals.

2.2. Methods

See Table 2 for methods used in the previous studies.

Hand preferences were assessed using the tube task [55]. For this task, the individual holds a tube with one hand while reaching for food inside with a finger of the other hand. In this bimanual coordination, the hand assuming the active role is the one reaching for food, and the hand holding the tube is subordinate. The tubes were white plastic tubes (15 cm length, 32 mm diameter) baited with some sticky food (e.g. honey, peanut butter, syrup). The food was smeared on the inside edge at both ends of the tubes (10 cm length, 3 mm thick). The experiment was conducted by one of the authors (AC), with the help of the zoo keepers. The baited tubes were given to the group in their home cage (spread on the floor when the bonobos were outside or thrown to them from outside the enclosure). The number of tubes was superior to the number of individuals (three extra) to avoid fights and allow low ranked individuals to have access to the tubes [55,120,124]. We did several test sessions to collect a sample of data that would be of appropriate size to statistically detect individual preferences [72,78,79]. Ten test sessions were performed for each bonobo group, at the rate of one test session per day for 10 days. The experiment was conducted in September 2005 at Twycross, June 2006 at Stuttgart and September 2006 at Apenheul. Test sessions were videotaped to allow thorough recording of the hand and finger use. Video analysis was done using Windows media player. Focal animal sampling [1] was used as sampling technique.

2.2.1. Data recorded

We recorded the individual's identity, which hand held the tube, which hand was used to remove the food from the tube, which digit was used (index, middle finger, thumb, ring finger, index and middle fingers, middle and ring fingers, unknown/digit not visible), the number of tubes held by the subordinate hand (1–4), the number of tubes held by the active hand (if any), the posture of the subject (e.g. seated, bipedal, supine) and specific comments (tube held by one foot or by another individual).

For the analyses we removed all cases when: the subject held more than one tube in the subordinate hand, held a tube in the active hand, held the tube with one foot and when the tube was held by another individual. This requirement ensured that we analyzed bimanual coordinated actions only, and that the behaviour studied was standardized between the individuals (all subjects holding one tube only).

2.2.2. Recording techniques

We used two different recording techniques: frequency and bouts.

Table 3
Subjects characteristics

Zoo	Name	Sex	Date of birth	Age class	Rearing	Mother
Ap	Jasiri	2	06 November 2002	juv	Parent	Lomela
Ap	Jill	2	15 July 1985	adult	Parent	
Ap	Kumbuka	2	09 July 1999	ado	Parent	
Ap	Liboso	2	17 January 1998	ado	Parent	Zuani
Ap	Lingala	2	17 July 2003	juv	Parent	Jill
Ap	Lomela	2	19 May 1992	adult	Parent	
Ap	Mwindu	1	1985	adult	Parent	
Ap	Zuani	2	1990	adult	Parent	
St	Banbo	2	03 September 2002	juv	Hand	Banja
St	Chipita	2	1993	adult	Hand	
St	Daniela	2	17 June 1968	adult	Parent	
St	David	1	27 July 2001	juv	Parent	Daniela
St	Diwani	1	11 August 1996	ado	Parent	Daniela
St	Khaya	2	19 October 2001	juv	Hand	Banja
St	Kirembo	1	10 December 1992	adult	Parent	
St	Kombote	2	1966	adult	Hand	
St	Lina	2	28 July 1985	adult	Parent	
St	Louisoko	1	19 April 1998	ado	Parent	Lina
St	Lucuma	1	29 October 2002	juv	Parent	Lina
St	Mixi	2	18 December 2001	juv	Parent	Chipita
St	Zorba	1	1980	adult	Hand	
Tw	Banja	2	01 February 1990	adult	Parent	
Tw	Cheka	2	18 March 1996	ado	Parent	
Tw	Diatou	2	21 October 1977	adult	Hand	
Tw	Jasongo	1	02 August 1990	adult	Hand	
Tw	Kakowet II	1	07 June 1980	adult	Hand	
Tw	Keke	1	02 January 1994	adult	Parent	Diatou
Tw	Kichele	2	19 April 1989	adult	Parent	Diatou
Tw	Luo	1	01 December 2002	juv	Parent	Diatou

Zoo: Apenheul (Ap), Twycross (Tw), Stuttgart (St); age class: juvenile (juv), adolescent (ado) adult; rearing: parent-reared (parent), human-reared (hand)

- *Frequency*: We counted every time that the subject "inserted its finger into the tube and brought it to the mouth". This is the recording technique that has been used in most other studies (Table 2), and that will be used here for the comparisons. However, this recording method has been criticized on the grounds of lack of data independency [79,91]. The use of one hand may influence the following use of this hand in the subsequent trials of a sequence. Counting every trials of a sequence could thus introduce biases in the sample of data [72,79].
- *Bouts*: To ensure that the data points were strictly independent of each others, we recorded bouts for the action "inserting the finger into the tube and bringing it to the mouth". We counted one bout for each sequence of identical actions, by recording only the first pattern of a sequence [72,78,79,124]. We counted a new bout when two sequences were separated by an action by which the subject could change its hand [79]. A new bout was considered only after the individual had dropped the tube, changed the hand holding the tube or manipulated the tube with both hands before eating again (locomotion with the tube in one hand was not considered a separating event).
- *First 50 frequency data points*: Frequency and bouts differed as regards to the mean number of data points per subject (272 with frequency, 47.62 with bouts). To investigate the possible effect of this sample size difference, we added a new variable that considered only the first 50 frequency data points (the number varied between 47 and 53 because we kept the last bout entire). This allowed comparison of bouts and frequency with more similar sample sizes (mean = 50).

2.3. Statistical analysis

A handedness index score (HI) was calculated for each subject using the formula: $(R-L)/(R+L)$ [92]. R and L represent the number of times that the right and left hands were used to remove the food from the tube. HI reveals the direction of the preference from −1 to +1, with negative values indicating a left hand preference, and positive values a right hand preference. The absolute value of HI (ABSHI) gives the strength of preference from 0 to 1. The binomial test (B) [105] was used to evaluate the significance of differences between the use of the left and right hands for each individual. The group-level bias was assessed using one sample T-test (T) [105] performed on the HI values, and the binomial test performed on the number of right-handed and left-handed individuals. The influence of different variables (zoo, sex, age, rearing history, number of data points) on laterality was tested using Mann–Whitney (MW), Kruskall–Wallis (KW), Wilcoxon (W) and Spearman correlation tests [105]. Lateral biases were considered significant for $p \leq 0.05$ (two-tailed). Analyses were carried out using SPSS 13.

3. Results

3.1. Data collected

Only the individuals that had at least 15 bouts were included in the analyses, which rendered 29 individuals. We had a total of 7889 data points for the frequency and 1381 bouts. The mean number of data points per subject was 272 (S.E. 5.68) (mini 47, maxi 656) for the frequency, and the mean number of bouts per subject was 47.62 (S.E. 0.71) (mini 15, maxi 98). The mean number of data points per bout was 5.71 (mini 1, maxi 44).

3.2. Effect of the number of data points per subject

The number of data points per subject was similar in all three categories of individuals (right-handed, left-handed, not lateralized) (KW frequency: $p = 0.43$, bouts: $p = 0.443$). The number of data points was not correlated to the HI values (Spearman frequency: rho = 0.071, $p = 0.716$, bouts: rho = −0.145, $p = 0.454$), nor to the ABSHI values (Spearman frequency: rho = 0.108, $p = 0.578$, bouts: rho = −0.205, $p = 0.286$).

3.3. Effect of the zoo

We compared the data between the three zoos (Twycross, Stuttgart, Apenheul). No significant effect of the zoo was found on HI values (KW frequency: $p = 0.498$, bouts: $p = 0.51$), on ABSHI values (KW frequency: $p = 0.445$, bouts: $p = 0.5$) and on the number of data points per subject (KW frequency: $p = 0.07$, bouts: $p = 0.097$). There was no significant difference between the zoos, and we thus analysed the combined data-set of the three zoos.

Table 4
Data with frequency

Zoo	Individuals	Right	Left	Total	HI	ABSHI	Binomial test	Category
Ap	Jasiri	65	296	361	−0.64	0.64	2E−36	L
Ap	Jill	651	5	656	0.985	0.985	7E−186	R
Ap	Kumbuka	32	156	188	−0.66	0.66	9E−21	L
Ap	Liboso	341	51	392	0.74	0.74	9E−54	R
Ap	Lingala	30	72	102	−0.412	0.412	4E−05	L
Ap	Lomela	112	432	544	−0.588	0.588	3E−45	L
Ap	Mwindu	154	85	239	0.289	0.289	1E−05	R
Ap	Zuani	170	450	620	−0.452	0.452	4E−30	L
St	Banbo	13	74	87	−0.701	0.701	2E−11	L
St	Chipita	86	118	204	−0.157	0.157	0.0297	L
St	Daniela	214	6	220	0.945	0.945	2E−55	R
St	David	8	177	185	−0.914	0.914	1E−42	L
St	Diwani	16	376	392	−0.918	0.918	2E−90	L
St	Khaya	74	102	176	−0.159	0.159	0.0415	L
St	Kirembo	167	6	173	0.931	0.931	6E−42	R
St	Kombote	101	149	250	−0.192	0.192	0.0029	L
St	Lina	51	200	251	−0.594	0.594	5E−22	L
St	Louisoko	132	5	137	0.927	0.927	4E−33	R
St	Lucuma	12	35	47	−0.489	0.489	0.0011	L
St	Mixi	69	52	121	0.14	0.14	0.1455	Not lat
St	Zorba	333	6	339	0.965	0.965	4E−90	R
Tw	Banja	385	218	603	0.277	0.277	1E−11	R
Tw	Cheka	106	130	236	−0.102	0.102	0.1342	Not lat
Tw	Diatou	61	212	273	−0.553	0.553	1E−20	L
Tw	Jasongo	146	0	146	1	1	2E−44	R
Tw	Kakowet II	124	117	241	0.029	0.029	0.6992	Not lat
Tw	Keke	328	15	343	0.913	0.913	7E−78	R
Tw	Kichele	208	92	300	0.387	0.387	2E−11	R
Tw	Luo	18	45	63	−0.429	0.429	0.0009	L
Total/mean		4207	3682	7889	0.0196 (S.E. 0.023)	0.5686 (S.E. 0.011)		

Number of right hand responses (right), number of left hand responses (left) and total number of responses (total). HI and ASBHI values. Binomial test: p value of the binomial test. Category: left-handed (L), right-handed (R), not lateralized (Not lat) individuals based on the binomial test (11 right-handed, 15 left-handed, 3 not lateralized subjects).

3.4. Frequency (number of responses per subject) (Table 4)

Amongst the 29 bonobos, 26 (89.66%) were lateralized (B $p \leq 0.042$) and 3 (10.34%) had no significant preference (B $p \geq 0.134$). The number of lateralized bonobos was greater than the number of not lateralized individuals (B $p < 0.001$). The mean ABSHI was 0.569 (S.E. 0.011). The mean percentage of use of the preferred hand was 81.18% (S.E. 0.544, N = 26). Amongst lateralized bonobos, 11 were right-handed and 15 were left-handed. The mean HI value was 0.02 (S.E. 0.023). There was no bias at the group-level regarding HI values (T $t = 0.160$, $p = 0.874$) and regarding the number of right-handed compared to left-handed subjects (B $p = 0.557$).

3.5. Bouts (number of bouts per subject) (Table 5)

Amongst the 29 bonobos, 17 (58.62%) were lateralized (B $p \leq 0.025$) and 12 (41.38%) had no significant preference (B $p \geq 0.053$). The number of lateralized subjects did not differ from the number of not lateralized individuals (B $p = 0.458$). The mean ABSHI was 0.492 (S.E. 0.01). The mean percentage of use of the preferred hand was 83,977 (S.E. 0.70, N = 17). Amongst lateralized bonobos, 8 were right-handed and 9 were left-handed. The mean HI value was 0.026 (S.E. 0.02). There was no bias at the group-level regarding HI values (T $t = 0.224$, $p = 0.824$), and regarding the number of right-handed compared to left-handed subjects (B $p = 1$).

3.6. Comparison frequency and bouts

There were significantly more lateralized subjects with frequency than bouts (B $p = 0.035$). The ABSHI values were higher with frequency (mean = 0.569) than with bouts (mean = 0.492) (W $z = -3.234$, $p = 0.001$). The HI values were similar with the two measures (W $z = -0.296$, $p = 0.767$). The HI values with frequency and bouts were correlated (Spearman rho = 0.972, $p < 0.001$) and so were the ABSHI values (Spearman rho 0.936, $p < 0.001$).

3.7. Bout length

The bout length is the number of frequency data points per bout. We compared the bout length between the right and left hands for each individual. For right-handed subjects, the mean length of bouts with the right hand (8.525) was greater than the length of bouts with the left hand (4.619) (W $z = -2.8$, $p = 0.005$). Similarly, for left-handed individuals the length of left hand bouts (5.56) was greater than the length of right hand bouts (3.84) (W $z = -2.726$, $p = 0.006$). For the three not lateralized individuals there was no difference between the right (4.531) and left hands (4.18) (W $z = 0$, $p = 1$) regarding bout length. The mean length of bouts performed with the preferred hand was similar for right-handed (4.619) and left-handed subjects (5.56) (MW, $p = 1$).

3.8. First 50 frequency data points (Table 6)

Frequency and bouts differed regarding the mean number of data points per subject (272 with frequency, 50 (47.62) with bouts). To investigate the effect of this sample size difference, we considered only the first 50 frequency data points. With the first 50 frequency data points, 22 (75.86%) bonobos were lateralized (B $p \leq 0.049$) and 7 (24.14%) had no significant preference (B $p \geq 0.119$). The number of lateralized bonobos was greater than the number of not lateralized individuals (B $p = 0.008$). The mean ABSHI was 0.583 (S.E. 0.012). The mean percentage of use of the preferred hand was 86.724 (S.E. 0.523, N = 22). Amongst lateralized bonobos, 9 were right-handed and 13 were left-handed. The mean HI value was −0.056 (S.E. 0.024). There was no bias at the group-level regard-

Table 5
Data with bouts

Zoo	Individuals	Right	Left	Total	HI	ABSHI	Binomial test	Category
Ap	Jasiri	17	60	77	−0.5584	0.5584	8.9E−07	L
Ap	Jill	66	1	67	0.97	0.97	9E−19	R
Ap	Kumbuka	15	41	56	−0.464	0.464	0.0007	L
Ap	Liboso	45	15	60	0.5	0.5	0.0001	R
Ap	Lingala	14	34	48	−0.417	0.417	0.0055	L
Ap	Lomela	23	42	65	−0.2923	0.2923	0.0248	L
Ap	Mwindu	17	16	33	0.03	0.03	1	Not lat
Ap	Zuani	17	34	51	−0.333	0.333	0.0241	L
St	Banbo	9	41	50	−0.64	0.64	6E−06	L
St	Chipita	21	33	54	−0.222	0.222	0.1337	Not lat
St	Daniela	14	1	15	0.867	0.867	0.001	R
St	David	4	37	41	−0.805	0.805	1E−07	L
St	Diwani	7	60	67	−0.791	0.791	1E−11	L
St	Khaya	27	43	70	−0.229	0.229	0.0722	Not lat
St	Kirembo	18	2	20	0.8	0.8	0.0004	R
St	Kombote	20	29	49	−0.184	0.184	0.2529	Not lat
St	Lina	7	17	24	−0.417	0.417	0.0639	Not lat
St	Louisoko	15	1	16	0.875	0.875	0.0005	R
St	Lucuma	6	16	22	−0.455	0.455	0.0525	Not lat
St	Mixi	21	15	36	0.167	0.167	0.405	Not lat
St	Zorba	46	1	47	0.957	0.957	7E−13	R
Tw	Banja	57	41	98	0.163	0.163	0.1293	Not lat
Tw	Cheka	31	24	55	0.127	0.127	0.4188	Not lat
Tw	Diatou	13	32	45	−0.422	0.422	0.0066	L
Tw	Jasongo	25	0	25	1	1	6E−08	R
Tw	Kakowet II	18	32	50	−0.28	0.28	0.0649	Not lat
Tw	Keke	53	4	57	0.86	0.86	6E−12	R
Tw	Kichele	40	27	67	0.194	0.194	0.142	Not lat
Tw	Luo	6	10	16	−0.25	0.25	0.4545	Not lat
Total/mean		672	709	1381	0.0259 (S.E. 0.02)	0.4921 (S.E. 0.01)		

Number of right hand responses (right), number of left hand responses (left) and total number of responses (total). HI and ASBHI values. Binomial test: *p* value of the binomial test. Category: left-handed (L), right-handed (R), not lateralized (Not lat) individuals based on the binomial test (8 right-handed, 9 left-handed, 12 not lateralized subjects).

Table 6
Data with first 50 frequency data points

Zoo	Individuals	Right	Left	Total	HI	ABSHI	Binomial test	Category
Ap	Jasiri	0	47	47	−1	1	1E−14	L
Ap	Jill	50	0	50	1	1	2E−15	R
Ap	Kumbuka	11	39	50	−0.56	0.56	9E−05	L
Ap	Liboso	31	19	50	0.24	0.24	0.119	Not lat
Ap	Lingala	13	36	49	−0.469	0.469	0.001	L
Ap	Lomela	23	26	49	−0.061	0.061	0.775	Not lat
Ap	Mwindu	25	25	50	0	0	1.112	Not lat
Ap	Zuani	7	44	51	−0.725	0.725	1E−07	L
St	Banbo	9	41	50	−0.64	0.64	6E−06	L
St	Chipita	25	28	53	−0.057	0.057	0.784	Not lat
St	Daniela	51	0	51	1	1	9E−16	R
St	David	0	52	52	−1	1	4E−16	L
St	Diwani	1	52	53	−0.962	0.962	1E−14	L
St	Khaya	18	33	51	−0.294	0.294	0.049	L
St	Kirembo	48	3	51	0.882	0.882	2E−11	R
St	Kombote	19	30	49	−0.224	0.224	0.152	Not lat
St	Lina	2	47	49	−0.918	0.918	4E−12	L
St	Louisoko	48	5	53	0.811	0.811	7E−10	R
St	Lucuma	12	35	47	−0.489	0.489	0.001	L
St	Mixi	25	23	48	0.042	0.042	0.885	Not lat
St	Zorba	45	6	51	0.765	0.765	2E−08	R
Tw	Banja	38	15	53	0.434	0.434	0.002	R
Tw	Cheka	21	27	48	−0.125	0.125	0.471	Not lat
Tw	Diatou	5	43	48	−0.792	0.792	1E−08	L
Tw	Jasongo	51	0	51	1	1	9E−16	R
Tw	Kakowet II	17	35	52	−0.346	0.346	0.018	L
Tw	Keke	45	4	49	0.837	0.837	8E−10	R
Tw	Kichele	40	9	49	0.633	0.633	9E−06	R
Tw	Luo	10	40	50	−0.6	0.6	2E−05	L
Total/mean		690	764	1454	−0.056 (S.E. 0.024)	0.583 (S.E. 0.012)		

Number of right hand responses (right), number of left hand responses (left) and total number of responses (total). HI and ASBHI values. Binomial test: *p* value of the binomial test. Category: left-handed (L), right-handed (R), not lateralized (Not lat) individuals based on the binomial test (9 right-handed, 13 left-handed, 7 not lateralized subjects).

ing HI values (T $t = -0.441$, $p = 0.662$) and regarding the number of right-handed compared to left-handed subjects (B $p = 0.523$).

3.9. Comparison bouts and first 50 frequency data points

The HI values with bouts (mean = 0.026) were higher than the HI values with the first 50 frequency data points (mean = −0.056) (W $z = -2.054$, $p = 0.04$). The ABSHI values were higher with the first 50 frequency data points (mean = 0.583) than with bouts (mean = 0.492) (W $z = -2.174$, $p = 0.03$). The HI values were correlated between the two measures (Spearman rho = 0.946, $p < 0.001$), and so were the ABSHI values (Spearman rho = 0.81, $p < 0.001$).

3.10. Comparison frequency and first 50 frequency data points

The HI values with frequency (mean = 0.02) were higher than the HI values with the first 50 frequency data points (mean = −0.056) (W $z = -2.095$, $p = 0.036$). The ABSHI values were similar with the two measures (W $z = -0.615$, $p = 0.539$). The HI values were correlated between the two measures (Spearman rho = 0.938, $p < 0.001$), and so were the ABSHI values (Spearman rho = 0.79, $p < 0.001$).

3.11. Finger use (based on frequency data)

Of the cases when the finger was visible, the index finger was used 84.66%, the thumb was used 6.9%, the middle finger 6.14% and other fingers or several fingers 2.3%. The use of several fingers together was specific to young individuals.

3.12. Strength of laterality

Right-handed individuals had stronger preferences than left-handed individuals (MW frequency: $p = 0.027$, bouts: $p = 0.002$) (mean ABSHI: frequency: 0.76 for right-handed, 0.524 for left-handed bonobos, bouts: 0.854 for right-handed, 0.518 for left-handed individuals). The number of data points per subject was similar for right-handed and left-handed individuals (MW frequency: $p = 0.299$, bouts: $p = 0.23$).

3.13. Effect of sex

We found no difference between males and females regarding HI values (MW frequency: $p = 0.301$, bouts: $p = 0.50$). With frequency, there was no sex difference regarding ABSHI values (MW frequency: $p = 0.08$) and the number of data points (MW frequency: $p = 0.144$). With bouts, there was a trend for males to be more strongly lateralized than females (MW bouts: $p = 0.053$) (mean ABSHI: 0.646 for males, 0.398 for females), and we had fewer data points per subject for males than for females (MW bouts: $p = 0.018$) (mean number: 35.82 for males, 54.83 for females).

3.14. Effect of age class (adults, adolescents, juveniles)

The direction of preference varied between age classes (KW: frequency: $p = 0.042$, bouts: $p = 0.034$). The HI values were significantly higher in adults compared to juveniles (MW frequency: $p = 0.012$, bouts: $p = 0.008$) (mean HI: frequency: −0.45 for juveniles, −0.003 for adolescents and 0.26 for adults, bouts: −0.398 for juveniles, 0.049 for adolescents, 0.231 for adults). With frequency (KW $p = 0.007$), but not with bouts (KW $p = 0.709$), the number of data points per subject varied between age categories. With frequency, we had fewer data points per subject for juveniles than for adolescents (MW frequency: $p = 0.028$) and adults (MW $p = 0.003$) (mean: 142.75 for juveniles, 269 for adolescents, 337.63 for adults). There was no difference between age classes regarding ABSHI values (KW frequency: $p = 0.565$, bouts: $p = 0.85$).

3.15. Effect of rearing history

We found no difference between mother-reared and human-reared bonobos regarding HI values (MW frequency: $p = 0.77$, bouts: $p = 0.884$), ABSHI values (MW frequency: $p = 0.464$, bouts: $p = 0.961$) and the number of data points per subject (MW frequency: $p = 0.435$, bouts: $p = 1$).

3.16. Effect of kinship

We compared the direction of preference in related pairs. Amongst mother–infant pairs, three pairs showed matched preferences between the mother and its infant (Lomela–Jasiri, Diatou–Luo, Lina–Lucuma) and nine pairs showed different preferences (Jill–Lingala, Zuani–Liboso, Banja–Banbo, Banja–Khaya, Diatou–Keke, Diatou–Kichele, Lina–Louisoko, Daniela–Diwani, Daniela–David). Regarding siblings, three pairs showed matched preferences (Banbo–Khaya, David–Diwani, Keke–Kichele) and three pairs showed different preferences (Louisoko–Lucuma, Keke–Luo, Kichele–Luo).

4. Discussion

We assessed hand preference in bonobos, a rarely studied species and outstanding model for investigating the evolution of handedness. We used a bimanual coordinated task that was complex and appropriate to reveal manual preference. This task has been used with other species, which allows reliable data comparisons. We recorded both frequency and bouts. The bonobos exhibited strong individual preferences. With frequency, 11 bonobos were classified as right-handed, 15 were left-handed and 3 had no preference. With bouts, 8 bonobos were right-handed, 9 were left-handed and 12 had no preference. With the first 50 frequency data points, 9 bonobos were right-handed, 13 were left-handed and 7 had no preference. There was no bias at the group-level. We will first consider variables that may have an influence on laterality. We analyze the effect of independent variables, such as sex, age, rearing history and kinship. We assess the effect of the different recording techniques by comparing the three variables (frequency, bouts, first 50 frequency data points). Then, we discuss the results in relation to previous data, and to theories on brain lateralization and handedness.

4.1. Individual variables

- The sex of the individual may influence laterality. Previous data are inconsistent [47,79], but there are some reports of an influence of sex on hand preference in non-human primates (e.g. [20,121]). In bonobos, we found no significant effect of sex on the direction or strength of laterality. We observed a trend for males to be more strongly lateralized than females (with bouts).
- Along with developmental processes, age may influence hand preference. Data are inconclusive, but some non-human primates studies have suggested that laterality may increase with age [79], with immature individuals exhibiting weaker or less consistent preferences than adults [42,54,55,86,119,123,124]. Here we found no effect of age on the strength of preference. Regarding the direction of preference, we found a greater use of the right hand in adults compared to juveniles (with frequency and bouts). This would go along with some previous bonobos data, for the behaviours feeding, reaching [44] and bimanual feeding (combined data from [44,48]). More data would be necessary

to determine the effects of age and sex on manual laterality in bonobos.

- Being raised by humans may have an influence on the hand preference of the subjects [61,79]. Data would indicate no significant effect of rearing on hand preference in great apes (Hopkins 'tube task' studies [55,56,63,64,66]) (but see [47] for rearing effects in apes). Here, we found no effect of rearing history: mother-reared bonobos and human-reared bonobos did not differ for the direction or strength of laterality. There are no previous data about rearing effect in bonobos.
- The transmission of hand preference is a controversial topic [25,30,47,79]. In chimpanzees, data for the 'tube task' would suggest a possible heritability of hand preference, but the mechanisms of transmission remain unclear [49,56,62]. To investigate heritability of hand preference, we compared the direction of preference in related pairs. Amongst mother–infant pairs, most pairs showed different preferences between the mother and its infant. Regarding siblings, there were equal numbers of pairs with matched preferences and different preferences. These are the first data on heritability of hand preference in bonobos. They would not go along with a transmission of preference from mother to offspring, but more data would be required to discuss this issue.
- Non-human primates studies generally found no difference in the strength of laterality between right-handers and left-handers (e.g. for the 'tube task': [104,120,124,125]). Here right-handed bonobos exhibited stronger preferences than left-handed individuals, suggesting a stronger lateralization in right-handers. This finding would match data in humans [25].
- Regarding finger use, the bonobos preferred to use their index finger to remove the food from the tube, which is similar to the other species behaviour [55,85,104,120,124].

4.2. Variables frequency, bouts, first 50 frequency data points

We used two different recording techniques: frequency and bouts. We recorded frequency, because this measure has been used in most other studies and was necessary for comparisons. However, recording frequency is criticized on the grounds of lack of data independency [79,91]. The data points that are in a sequence of actions are not independent of each others. So recording every trial in a sequence may introduce biases in the sample of data [6,70,72,79,90,91]. Bouts are designed to record only data points that are independent of each others. They meet the data independency criteria, a formal requirement in laterality research [6,72,79]. Recording bouts might be discussed regarding differences in bout lengths [53]. We count one bout for one sequence of identical actions, but the sequences vary in length. Here, the number of data points per bout varied from 1 to 44 (mean = 5.71). We performed analyses on bout length, and found that right-handed bonobos showed longer bouts with the right hand, and the reverse for left-handed subjects. Thus asymmetries in bout length would be related to laterality: bouts with the preferred hand would be longer than bouts with the non-preferred hand. This result matches the one of Hopkins et al. [53] and indicates that bout length may be an interesting variable to consider *per se*.

There is a discussion regarding the use of bouts and frequency, and some efforts to demonstrate that frequency and bouts would provide similar results [53,60,61,64,66,90,91,124]. We compared here the two variables. For the strength of laterality, there was a difference between frequency and bouts. Laterality was stronger with frequency than with bouts: the ABSHI values were higher with frequency, and there were more lateralized individuals with frequency (90%) than with bouts (59%). This finding goes along with data in chimpanzees: great number of unlateralized subjects with bouts (46% [53], 64% [66]), and higher than with frequency (70 versus 22 unlateralized individuals [53]). For the direction of preference, the two measures gave quite similar results. Both the HI and ABSHI values were correlated between the two measures, which matches data in chimpanzees [53]. Therefore, our findings would add to previous data to suggest that the results could depend on the recording technique: laterality would be stronger with frequency than with bouts.

This difference might be related to the difference in sample size, as the sample was 5.71 times larger with frequency (mean = 272) than with bouts (mean = 47.62). Significant biases would be more likely to appear with big sample sizes as the binomial test is sensitive to the size of the sample [47,53,66,79]. To investigate the effect of this sample size difference, we considered only the first 50 frequency data points. This allowed comparisons of bouts and frequency with more similar sample sizes (mean = 50). Both the HI and ABSHI values differed between bouts and first 50 frequency data points. Laterality was stronger with first 50 frequency data points. This suggests that the difference between bouts and frequency regarding the strength of laterality would not be only a matter of sample size.

To further assess the effect of the sample size on laterality, we compared frequency (mean = 272) and first 50 frequency data points (mean = 50). The HI values differed between the two measures, and the ABSHI values were similar. Other analyses could also suggest a possible influence of the number of data points on laterality. Males tended to be more lateralized than females, and they had fewer data points than females (with bouts). Adults had higher HI values than juveniles (with frequency and bouts), and they had more data points than juveniles (with frequency). Another related field to investigate was the between subjects difference in the number of data points [90,92]. The number of data points per subject varied between 47 and 656 with frequency, and between 15 and 98 with bouts (no variation with first 50 frequency data points). This inter-subject variability is a common issue of studies on hand preference (e.g. [55,61,120]) (Table 2). Its influence on laterality has been discussed in terms of possible introduction of sampling biases [60,61,79,90,91]. We found no significant effect of the number of data points per subject on laterality, which would match data of other 'tube task' studies [55,120,124,125]. Our findings could suggest a possible influence of the sample size on laterality. The effect would be unclear, but this factor should not be completely ruled out.

4.3. Interpretation of the results in relation to previous findings and theories

4.3.1. Strength of preference and task complexity

For the 'tube task', the majority of the bonobos were lateralized (between 59% and 90% depending on the measure). The individual preferences were well marked, with a high percentage of use of the preferred hand (mean percentage: 81%, 84% and 87% depending on the measure). These findings match data of other species tested on this complex 'tube task' (Table 2, Fig. 1) and would indicate brain lateralization for manual functions in bonobos.

For these bonobos, we have also collected data for spontaneous daily activities (e.g. feeding, picking up, carrying item) [12]. The laterality for the 'tube task' was more pronounced than the laterality observed for spontaneous activities [12]. This result matches other species comparisons data between the 'tube task' and simpler tasks [85,95,104,120]. Therefore, our results add to previous 'tube task' data to indicate that this measure would reveal particularly strong laterality. Our findings match and append to the large database showing that laterality would be stronger in complex than sim-

ple tasks (e.g. [2,6,15,26,35,95,104,109,114,118]). This would support the "theory of task complexity" [26] regarding strong laterality in complex tasks.

4.3.2. Individual versus group-level preferences

For the 'tube task', the bonobos exhibited individual preferences. No group-level bias appeared. This task has revealed group-level biases in chimpanzees (review in [61]), orang-utans [63], infant rhesus macaques [125] and baboons [120], while individual-level preferences have been found in gorillas [63], white-faced capuchins [85], tufted capuchins [95,124], rhesus macaques [124] and De Brazza's monkeys [104] (Table 2, Fig. 1).

One could argue that differences between studies might be related to some methodological differences. We used exactly the same task to measure laterality in order to allow reliable data comparisons. However, there were still some methodological points that differed between studies, such as differences in the sample size of subjects, of data points per subjects, of test sessions (Table 2). As discussed above, such differences might have an influence on the results [79]. For instance, our sample ($N = 29$) was smaller than those of chimpanzees ($N \geq 109$) and baboons ($N = 104$), which would make a group-level bias more difficult to reveal here [66]. Group-level biases have nonetheless been reported in orang-utans and infant rhesus for smaller samples ($N = 19$), and in a subsample of chimpanzees ($N = 46$) [63]. Given the ratio of right- and left-handed individuals, it is unlikely that the absence of group-level bias in bonobos could be related to the size of our sample.

4.3.3. Evolutionary theories

Between species differences in laterality might be related to differences in the species characteristics. Brain lateralization and handedness would have evolved in relation to some specific selective pressures. Factors such as postural demands [84], tool use [68], throwing [9], bipedalism, gestural communication and language [18,40,117], complex manipulations [6], precise actions, bimanual coordination [26] have been proposed as mechanisms fostering the evolution of hemispheric specialization and handedness. Therefore, differences in biological features like the diet (foraging, food processing), the degree of arboreality and the locomotor behaviour, the social behaviour and the communicative skills may account for between species differences in laterality [47,79].

Along with evolutionary theories and the special characteristics of bonobos, the bonobos brain lateralization should be close to the one of humans. Bonobos exhibit linguistic abilities that could surpass those of the other apes [102,103]. Both language and handedness are lateralized in the left hemisphere in humans [25], and the evolution of these two features could be related [18,40,117]. With their highly evolved language skills, bonobos could display left hemisphere specialization for language and manual functions, and would be expected to exhibit a group-level right bias. Our findings would not support this hypothesis. Bonobos would be the non-human primates that most frequently use bipedal locomotion [24,112,113]. Bipedalism has been proposed to be related to the emergence of right-handedness in humans. Along with this theory, bonobos would be expected to show the highest right-hand use and a group-level right-bias. Our results would not support this hypothesis.

To investigate the evolution of human brain lateralization, data on hand preference in great apes are crucial. As seen in the review of findings (Table 1), great apes have been shown to exhibit hand preferences. The preferences appeared especially in complex tasks that require high cognitive skills (e.g. bimanual coordinated actions, tool use). Group-level biases have been found only for certain behaviours (e.g. bimanual feeding, 'tube task', bipedal reaching, gesturing), and they were not very marked (65%).

Amongst the apes, *Pan* is the closest genus to humans [69], which makes chimpanzees and bonobos the best candidates to exhibit precursors of human features. Chimpanzees have been reported to exhibit a group-level right bias for several tasks, including the 'tube task' (Tables 1 and 2). However, these findings have also been disputed [79,90,91].

Bonobos exhibited strong preferences for the 'tube task'. This result matches the previous bonobos data on hand use in complex experimental tasks [14,16,23,44]. This would indicate that the brain of bonobos may be lateralized for manual functions. Our findings would go along with the idea that brain lateralization may have been selected to improve brain and behavioural efficiency for complex processes, such as bimanual coordination. For the 'tube task', the bonobos exhibited individual preferences, but no group-level bias appeared. This finding matches the previous bonobos data [37,44,48,107] (except group-level biases reported for initiating locomotion (right bias) and for carrying (left bias) with combined data from [44,48]).

To interpret the findings in relation to evolutionary theories, one must take into account the methodological issues that hinder research on laterality. Findings generally reported in non-human primates are: weak and inconsistent preferences; scarce and disputed evidence of group-level biases. Considering the issue of task complexity, the lack of evidence for human-like laterality in non-human primates might be partly related to the use of tasks that would be too simple and inappropriate to reveal laterality [6,61,72]. Considering sample sizes issues, the absence of group-level bias in non-human primates might be related to the small samples considered ($N < 10$ in most studies [47,72]).

There is strong evidence of hand preferences in non-human primates. The fact that laterality varies according to the task would indicate that the expression of brain lateralization may be more sensitive to influential factors (e.g. position of the object relative to the hand [47,122], complexity of the task [26], posture of the subject [84,101]). The available database would suggest that the difference between humans and non-human primates may be in degree rather than nature [17]: a weaker form of lateralization in non-human primates. A precursor of human brain lateralization may have existed in a common ancestor of humans and apes several million years ago, but the stronger lateralization and handedness observed in humans would be a derived trait, specific to humans and evolved in the hominin lineage after the divergence from the last common ancestor [6,79].

4.4. Perspectives

For future research, we would like to emphasize the need to consider large samples of subjects, to use tasks that would be appropriate to reveal laterality, and to standardize the methods between studies to allow data comparisons between studies and species [61]. The bimanual coordination associated with the precise action seems to make the 'tube task' complex enough to require and evoke preferences in hand use. So, in further work, we will assess hand preferences with the 'tube task' in a larger sample of bonobos (Lola ya bonobo sanctuary, Democratic Republic of Congo, $N = 57$).

We deplore the disappointing lack of comparability between studies on humans and studies on non-human primates, which might partly explain some difference in the results. We will thus assess hand preferences in humans with the 'tube task' (in collaboration with J. Fagard). This will be one of the first times that exactly the same task is used with humans and non-human primates (see [126]). It would provide invaluable data for direct comparisons

between humans and other primates, allowing reliable assessment of differences in manual laterality.

Acknowledgements

We would like to acknowledge the following persons for their help and support. We warmly thank the keepers for their help with the management of the tubes and the bonobos (Cathy Hugues, Lily Hugues and Liz Cubberley from Twycross zoo, Beatrix Jarczewski, Ralf Schirsching, Andreas Krauss, Maik Denli, Tobias Weigold, Stefan Plaum and Sabine Kellewald from Stuttgart Wilhelma, Jacqueline Ruijs, Ria Bakker, Frank Rijsmus, Adrienne Polderman and Iris Schapelhoutman from Apenheul). We thank the research coordinators of these zoos for permission to conduct this experimental work (Sara Smith, Joanna Buerling and Jackie Hooley from Twycross zoo, Marianne Holtkötter from Stuttgart Wilhelma, Franck Rietkerk from Apenheul). We thank Catherine Blois-Heulin, Cédric Guillaume and anonymous reviewers for their advices on this article. This work is part of A. Chapelain PhD research, funded by Loughborough University.

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